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Case study: a hybrid plant in Hat Yai that produces more than enough energy for its shop

18 October 2026 · Rev07 · Prepared by Solar ESS from remote monitoring data and inverter settings. Customer details removed.

Summary

The plant is working well. Its 10 × 580 W panels (5.8 kWp) yield about 4.1 kWh per kWp per day, which is good for Hat Yai, and the system produces more than enough energy for the shop's consumption. Since the battery was added on 18 May 2026, the shop has covered 98% of its own electricity.

The inverter settings checked on 18 Oct 2026 suit the site. The points worth attention are the battery's energy readings, the inverter's operating temperature, using the midday surplus, and completing the monitoring records.

Plant at a glance

Review of all monitoring data from commissioning on 29 Jan 2026 to 17 Oct 2026 (262 days), from SolisCloud.

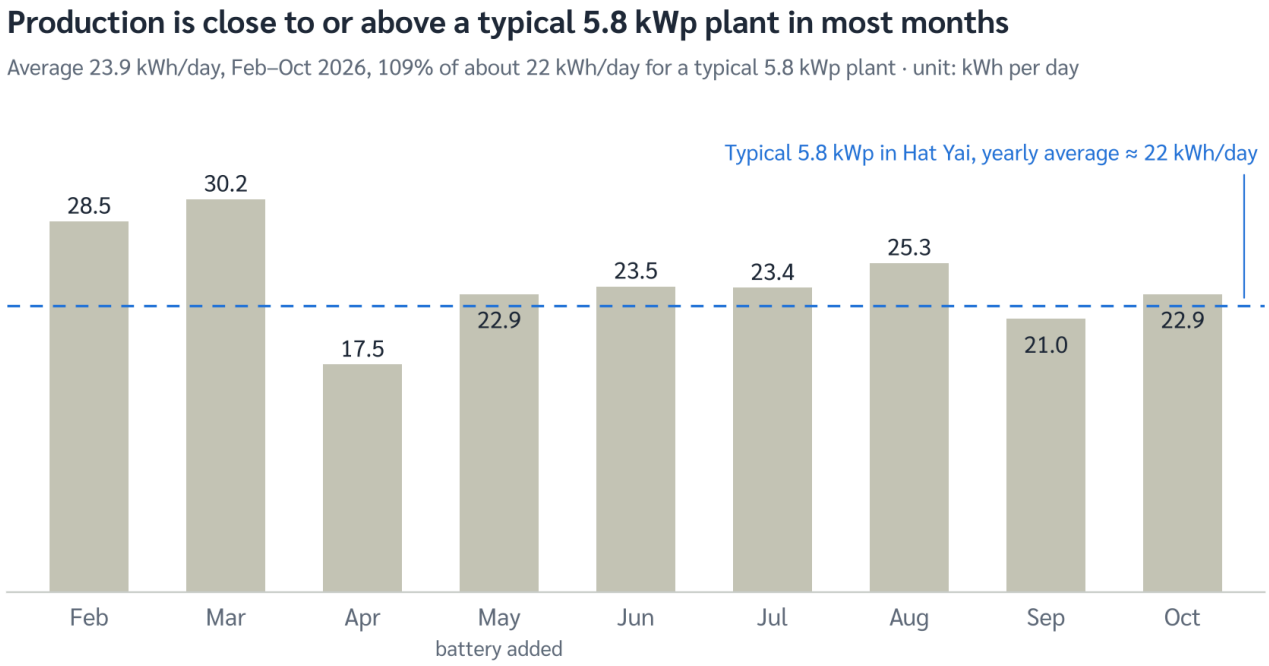
Item	Value
Location	Hat Yai, Songkhla
PV array	10 × 580 W panels in one string on MPPT1 = 5.8 kWp
Inverter	Solis S6-EH3P10K hybrid, 10 kW, three-phase; MPPT2 unused
Battery	One 48 V battery, added 18 May 2026
Total production	6,230 kWh (about 4.1 kWh per kWp per day)
Best day	36 kWh (25 Mar 2026), about 6.2 kWh per kWp
Total consumption	6,183 kWh (about 22–23 kWh/day)
Self-sufficiency since battery	98% (grid purchase 0.5 kWh/day, was 12.4)
Active alarms	None

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Inverter settings checked on 18 Oct 2026

Setting	Value	Comment
Storage mode	Self Use	Correct for this site
Unbalance Output	ON	Good: the shop's load sits mainly on L1 and L3
Allow Grid Charge	Enable	The battery may charge from the grid
Grid code	PEA	Correct for Thailand (read from the inverter)

Monthly performance



Source: SolisCloud daily totals, Feb–Oct 2026 (Oct = 1–17 Oct). The typical value is an approximate yearly average.

Production is close to or above a typical 5.8 kWp plant in most months, and well above it in the dry season. On sunny days the battery is full by early afternoon.

Findings

1. More energy than the shop uses

On sunny days the battery is full by early afternoon and the panels produce about 3 kWh a day more than the shop uses. Moving heavy loads to midday would put that surplus to work.

2. Battery energy counters do not add up

Since May the battery shows 1,832 kWh discharged but only 1,526 kWh charged. A battery cannot give out more than it takes in, so its readings are wrong. SolisCloud lists the battery maker as "OTHER". If the inverter misreads the battery's charge level, the battery can be over-discharged or never fully charged.

3. Steady overnight load

The shop draws about 0.7 kW all night, every night: about 8–9 kWh, close to 40% of daily use. Every 0.3 kW cut saves about 2.6 kWh of battery each night.

4. Inverter runs hot

The inverter's internal temperature was 58 °C at 2 kW and 58.3 °C at 3.3 kW (18 Oct 2026, 11:37 and 13:30). Inverters reduce output when they get too hot.

5. Monitoring records are incomplete

No purchase tariff is set and every kWh produced is valued at a flat 4.7 THB, so the earnings shown in the app are not meaningful. The number and wattage of the panels are not entered either.

Recommended actions

Priority	Action	Expected benefit
1	Move heavy loads to midday, once the battery is full	About 3 kWh a day more used on site
2	Check the battery brand setting and the battery-to-inverter communication; recalibrate the charge level	Correct charge readings and longer battery life
3	List what runs overnight and switch off what isn't needed	About 2.6 kWh of battery saved per night for each 0.3 kW cut
4	Check the inverter's ventilation and shading from direct sun	Avoids output cuts on hot days
5	Enter the purchase tariff and the panel data (10 × 580 W) in SolisCloud	Correct earnings and performance figures

Notes on the data

- Source: SolisCloud daily totals since 29 Jan 2026, 5-minute inverter data for 1, 17 and 18 Oct 2026, and the inverter settings as shown in SolisCloud on 18 Oct 2026. No settings were changed.
- Array: 10 × 580 W panels, as advised. The logged string voltage (about 360 V at midday) is a little lower than usual for 10 such panels, so an open-circuit voltage check on site (about 520 V at 25 °C) is worth doing. Typical values (about 22 kWh/day for 5.8 kWp) are approximate yearly averages for southern Thailand, not a site-specific simulation.
- Prepared by Solar ESS from remote monitoring only; findings 2 and 4, and the string voltage, should be confirmed on site.